



DATABASE

ASSIGNMENT 2

Individual - 10 %

This assignment has been designed to make you review Chapters 1-6, focusing on more advanced SELECT commands.

COMP 2003 Relational Database

Assignment 2 - Relational Database

This assignment requires you to create, populate, and manipulate a database using SQL, then use advanced selection queries to extract relevant data. Your mark in this assignment counts for 10% of your final grade.

Assignment Requirements:

This is an individual assignment. All submitted work is to be your own work. Please ensure that you are familiar with the academic integrity policies referred to in the course syllabus.

Please submit your work as a Word document (docx - not pdf) as well as your finalized SQL script (sql).

Report formatting

- Name your report file *YourName_Assignment2.docx* • Give your report an appropriate title with separate title page.
- Please include your name and student number.
- For each question, please include the question's number and text.
- Please include the text of your queries (including comments, and a screenshot which clearly shows the execution of your commands and the results. You might need to send more than one screenshot to show the results for each question).

SQL formatting

- Name your SQL script *YourName_Assignment2.sql* and complete all your SQL work in this file. Inside I should find all your SQL commands required to complete the assignment.
- Please include your name and student number as a comment header at the start of the script.
- The data provided (customers.sql) is solely for the purpose of populating your database with data – do not complete your SQL work in this file.
- Utilize the SQL Standards document provided in week 2 – with note to capitalization of keywords and lower-case database, table and field names.
- Complex statements should be broken over several lines as appropriate.
- Remember to add a brief descriptive comment for each query. You do not need to document every single line, but you must demonstrate that you understand what the SQL is doing.
- Please write in complete sentences in the commenting for relevant questions.

Submitting your assignment

- Create a zip file containing your SQL script and report document.
- Submit assignment zip file via the course website -> Assignments. Do not submit via email.

Evaluation Method

For each question you will receive the following marks:

- Execution: A variable number of marks will be assigned whether the command will run as provided.
- Accuracy: A variable number of marks will be assigned whether the question is answered correctly.

Additionally, the following marks will be assigned to the SQL script as a whole

- Structure: Up to 7 marks will be assigned for well-structured scripts that follow SQL standards. i.e. capitalized commands, lower case field names with underscores where needed, new lines, etc.
- Commenting: Up to 4 marks will be assigned for helpful, descriptive comments to be included throughout the script

If you have any questions, please do not hesitate to ask me.

Good luck.

Questions

Use the Customers.sql file to populate your assignment database.

1. Make a database and name it 'Assignment2' and add the data from Customers.sql to your database.
Execution: 1.5 marks
Accuracy: 0.5 mark
2. Use a single line SQL command to return the structure of the table from Q1. In your report, tell me what command you would use to generate a list of all the tables in your database.
Execution: 1 mark
Written answer: 1 mark
3. Add a column named *bonus* to your table. Place it after *surname*. Enable the users to store bonuses with values such as: 1099.99 or 2.00 by choosing the proper data type for *bonus*. In your report, mention the data type you chose and another that you considered but chose not to use.
Execution: 1 mark
Accuracy: 2 marks
Written answer: 1 mark
4. Using a CASE command fill the data for the bonus column.
 - If customers hold a Mastercard and they live in Ontario the bonus will be 200.00
 - Otherwise, if they live in any other provinces and have Mastercard, they will get 155.99 CAD as bonus.
 - Then, all Visa holders will get 100.00 CAD
 - Then, customers with any other credit cards will get 55.99 CAD as bonus.
 - Lastly, non-credit card holders will receive a bonus of zero.
 In your report, explain the considerations that were made in constructing this statement.
Execution: 1 mark
Accuracy: 5 marks
Written answer: 1 mark

5. Let's increase the bonuses. Please increase all the bonuses by 320.00 CAD for customers who are working as a model or have the first name 'Mary'. But, please only apply this increase to the people who were only getting a \$155.99 bonus. (Hint: only 7 rows must be updated.)

Execution: 1 mark

Accuracy: 4 marks

6. We need to calculate the total dollar value of bonuses being given to customers in each province.

- List each province and tabulate the total value of bonus dollars provided to customers in that province and name the new column as *total_bonuses*.
- Sort the list by the total amount of bonuses provided in each province.

In your report, explain the considerations that were made in constructing this statement.

Execution: 1 mark

Accuracy: 3 marks

Written answer: 1 mark

7. Generate a list displaying *only* the five longest surnames in customers' table. In two columns display the surname, and the length of their surname in a field called *name_length*. Please have the list sorted from longest to shortest. In your comments, how you would return surnames 6 through 10 of this same list. An example table is provided with surnames 1 through 5, your results should mirror what is presented below.

surname	name_length
Hollingsworth	13

Blankenship	11
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Brockington	11
-------------	----

Hershberger	11
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McLaughlin	10
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Execution: 1 mark

Accuracy: 3 marks

Written answer: 1 mark

8. Write a query to show customers' full names, street address and city.

- Please return their first name in all capital letters.

Is it possible to show both their first and last name in one column in the returned selection?

- If it is, please alter your script to name the column 'full_name' and include both their first and last name in this column.
- If it is not possible, please explain why it isn't in your report.

Execution: 1 mark

Accuracy: (2|3) marks

Written Answer: (0|1) marks

9. Write a query to show how many customers have MasterCards and how many are using Visa. In your comments, explain how you would sort this table alphabetically by credit card type. An example table is provided below; these numbers are fictitious.

cctype	num_customers
Visa	589
MC	432

Execution: 1 mark

Accuracy: 3 marks

Written Answer: 1 mark

10. If you were asked to add a *total_bonuses* column to the output results of Q9's query tabulating the total amount of bonuses provided to Visa vs MC customers,? How you would do so?

Written Answer: 3 marks